

Ad Targeting Optimization: Approach Document (Unsupervised ML + Deep Learning)

This document provides a structured approach for navigating the **Ad Targeting Optimization** case study, which focuses on leveraging Unsupervised Machine Learning and Deep Learning to segment users for improved ad performance.

Case Study Overview

The core challenge of this case study is to transition from a generic advertising strategy to a **segmented, data-driven targeting system**. Using unsupervised techniques, the objective is to analyze user demographic and behavioral data to **discover latent groups** (clusters) that exhibit similar characteristics and, crucially, different ad-clicking behaviors. The insights derived from these clusters will be used to recommend optimized ad budget allocation.

Datasets

You will be primarily working with one integrated dataset:

- **ad_click_dataset.csv**: Contains user metadata, browsing context, and the binary click outcome.

Critical Constraint

The primary unsupervised model training must **NOT** use the target column, **click**. This column is reserved strictly for **external evaluation** (measuring the Click-Through Rate per cluster) after the clustering models have been built.

General Approach & Data Preparation

Unsupervised models, especially K-Means, are highly sensitive to data scale and dimensionality. A robust preprocessing pipeline is mandatory to ensure meaningful clusters are formed.

4. TASK 1 — Data Preprocessing Hints

Step	Purpose & Hint
Remove Useless Columns (4.1)	Drop id and full_name immediately, as they are non-predictive identifiers.
Handle Missing Values (4.2)	Hint: Given the mixed data types, consider mode imputation for categorical columns (gender, device_type, browsing_history, time_of_day) and mean or median imputation for the numerical column (age).
Encode Categorical Features (4.3)	Clustering requires numerical inputs. Hint: Use One-Hot Encoding for all categorical columns (e.g., gender, device_type, ad_position, browsing_history, time_of_day).
Scale Numerical Columns (4.4)	K-Means uses Euclidean distance, which is heavily influenced by features with larger scales. Hint: Apply StandardScaler or MinMaxScaler to the age column and all newly encoded/derived numerical features.
Create Additional Features (4.5)	Creating features like is_mobile_user (from device_type) can sometimes simplify the feature space and improve interpretability.

5. TASK 2 — Machine Learning Clustering (K-Means)

This task establishes a baseline clustering performance using the classic K-Means algorithm.

Step	Objective & Hint
Choose Optimal K	Hint: Iterate through a range of cluster numbers (e.g., 3 to 8). Plot the Elbow Method (Sum of Squared Errors) and calculate the Silhouette Score for each K. The optimal K should be chosen based on a balance of both metrics.
Train and Assign	Train the final K-Means model with the optimal K and assign cluster labels to every user in the dataset.

Visualize Clusters	Hint: Use Principal Component Analysis (PCA) to reduce the high-dimensional feature set down to 2 components. Plot the clusters in this 2D space for visual confirmation of separation.
Interpret Clusters	Hint: Calculate the average value for every preprocessed feature within each cluster. A heatmap visualizing these averages is the most effective way to profile each cluster (e.g., Cluster X is high in 'Mobile' and high in 'Shopping').

6. TASK 3 — Deep Learning Clustering (Autoencoder + K-Means)

This task uses a Deep Learning approach to create a lower-dimensional, more robust feature representation before clustering.

Step	Objective & Hint
Autoencoder (AE) Architecture (6.1 & 6.2)	The Autoencoder learns a compressed representation of the data. Hint: The Latent Layer (Bottleneck) size of 8 neurons is critical. The model learns to reconstruct the input, forcing the bottleneck to capture only the most essential features of the user profile. Train the model until the reconstruction loss stabilizes.
Extract Latent Features (6.3)	After training, use only the Encoder part of the model to transform the original dataset into the new 8-dimensional latent feature space.
K-Means on Latent Space (6.4)	Apply K-Means clustering to these 8-dimensional latent features. Hint: Use the same optimal K value identified in Task 2 for a direct comparison, but recalculate the Silhouette Score and other metrics to confirm if a different K is better in the latent space.

7. TASK 4 — Compare Machine Learning vs Deep Learning

This task systematically compares the quality of the clusters generated by the two approaches.

Internal Evaluation (Unsupervised Metrics)

Compare the **Silhouette Score** and **Davies-Bouldin Index** (DB Index) for both the ML-KMeans and DL-Autoencoder + K-Means models. A **higher Silhouette Score** (closer to 1) and a **lower DB Index** (closer to 0) indicate better, more compact, and distinct clusters.

External Evaluation (Business Metric: CTR)

This is the most critical business evaluation. Using the un-trained **click** column:

1. Calculate the Click-Through Rate (CTR) for every cluster in both models:

$$\text{CTR_cluster} = \text{Number of clicks in cluster} / \text{Total users in cluster}$$

2. Compare the results: The superior clustering model is generally the one that yields the **largest disparity** between the maximum and minimum CTR across its segments (i.e., it successfully isolated the high-value clickers from the low-value non-clickers).

Metric	ML (K-Means)	DL (Autoencoder + KMeans)
Silhouette Score	[Calculated Value]	[Calculated Value]
DB Index	[Calculated Value]	[Calculated Value]
Max CTR Cluster	[Calculated Value]	[Calculated Value]
Min CTR Cluster	[Calculated Value]	[Calculated Value]

8. TASK 5 & 9: Interpretation and Recommendations

The final step is to translate technical findings into actionable business strategy.

Cluster Interpretation (Task 5)

For the **better-performing model** (likely the DL approach), interpret at least **four clusters**. Profile each cluster by describing the dominant characteristics of its users (e.g., "Cluster 3 is characterized by young, mobile users, primarily browsing Social Media in the Morning").

Final Recommendations (Task 9)

Based on the calculated CTR for each cluster, provide concrete strategic advice:

- **High-Value Clusters:** Identify clusters with significantly **above-average CTR**. Recommend an **increased ad budget** and more aggressive targeting for these segments.
- **Low-Value Clusters:** Identify clusters with significantly **below-average CTR**. Recommend **reduced or zero ad budget** and potentially shifting to cheaper ad inventory for these segments.
- **Strategic Improvement:** Explain precisely *how* segmentation improves the overall predicted CTR and how the company can use these cluster profiles to develop tailored ad creatives.